Venkata Naga Sai Prabath Devalla

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EDUCATION

Bachelor of Technology 2022 – 2025
SRK Institute Of Technology Vijayawada
Electrical & Electronics Engineering

SKILLS

Python
Python, NumPy, Pandas, Scikit-learn, Data Cleaning, Data Manipulation, Exploratory Data Analysis (EDA), Scikit-learn, Matplotlib, Seaborn, Object-Oriented Programming.

MySQL
MySQL, SQL, JOINs, Subqueries, Commands, Common Table Expressions (CTEs), Window Functions, Indexing, CRUD Operations, Entity-Relationship (ER), Stored Procedures, Triggers, ACID Properties.

Advanced Statistics
Probability, Statistical Inference, Hypothesis Testing, Confidence Intervals, Central Limit Theorem, Sampling Techniques, Probability Distributions,, ANOVA, Chi-Square Tests.

Power BI
Data Modeling, DAX, Power Query, KPIDashboard Development, Drill-through Analysis, Row-Level Security (RLS), Data Transformation.

Machine Learning
Supervised Learning, Classification & Regression Algorithms (Linear Models, Distance based Models, Tree Models), Model Evaluation Metrics(Accuracy, Precision, Recall), Unsupervised Learning, Clustering, Overfitting & Underfitting, Cross-Validation, Hyperparameter Tuning.

Deep Learning
Deep Learning, Artificial Neural Networks (ANN), Convolutional Neural Networks (CNN), Recurrent Neural Networks (RNN), Long Short-Term Memory (LSTM), PyTorch, TensorFlow, Keras, Transfer Learning, Gradient Descent, Backpropagation, Optimization Algorithms (Adam, SGD), Regularization (Dropout, Batch Normalization), Computer Vision, OpenCV

Generative AI
LLMs, Prompt Engineering, RAG, Embeddings, Chunking Strategies, Tokenization, Context Windowing, Prompt Template, Reranking (Basics), LLM Evaluation (Qualitative + basic metrics), Agentic AI Concepts (Tools, Memory, Planning).

PROJECTS

GenAI-Powered Technical Query Analyzer
- Developed a GenAI-powered assistant application using Gemini LLM, LangChain, and Streamlit to classify user queries across 8+ technical domains including Python, SQL, ML, DL, LLMs, and Agentic AI.
- Designed a prompt engineering pipeline with structured JSON-based response generation, improving response consistency and enabling seamless UI integration for real-time query analysis.
- Implemented automated keyword extraction and contextual explanation generation to produce 3-4 technical keywords and concise 100-200 word summaries for every user query.
- Integrated dynamic Python code generation and curated learning resource recommendations, supporting technical interview preparation and self-learning workflows for multiple AI/DS topics.

Lung Cancer Detection using Convolutional Neural Network (CNN)
- Developed a Deep Learning-based CNN model using TensorFlow and Keras to classify lung cancer images into binary classes with 85-95% validation accuracy.
- Processed and augmented 1000+ medical images (224x224 resolution) using normalization, resizing, and augmentation techniques to improve model generalization and reduce overfitting.

- Designed a multi-layer CNN architecture with 3+ convolutional layers, pooling, dropout, and dense layers for efficient medical image classification.
- Improved model performance by 10-15% through hyperparameter tuning, dropout regularization (0.25-0.5), and optimization algorithms such as Adam and SGD.

Swiggy Delivery Time Prediction
- Developed a Machine Learning regression model using 45,000+ delivery records to predict food delivery ETA based on distance, traffic conditions, and rider-related features.
- Performed feature engineering using geospatial distance calculations, operational metrics, and time-based variables, improving prediction efficiency and model performance.
- Compared and evaluated 6+ regression algorithms including Linear Regression, Random Forest, and XGBoost, achieving the best performance with XGBoost (R2 ≈ 0.74, MSE ≈ 23.27).
- Built a real-time prediction workflow using Python, Pandas, and Scikit-learn, helping improve delivery efficiency and customer experience through faster ETA estimation.

Olist E-Commerce Data Analysis
- Analyzed 500K+ e-commerce records including orders, customers, payments, and products using SQL and Python-based data analysis techniques.
- Computed $16M+ Gross Merchandise Value (GMV) and generated 20+ business KPIs for customer behavior, operational performance, and sales trend analysis.
- Applied advanced SQL concepts including JOINs, CTEs, aggregations, subqueries, and Window Functions to extract actionable business insights from 9 relational tables.
- Identified top-performing regions and product categories, revealing that the top 3 regions contributed nearly 35% of total revenue and top categories generated 40%+ of orders.

Analyzing Hiring Trends and Employee Satisfaction
- Analyzed workforce data from 1000+ companies across multiple industries to study hiring patterns, employee satisfaction, and workforce distribution trends.
- Evaluated 5+ organizational factors including job security, work-life balance, promotion rate, and career growth opportunities using statistical analysis techniques.
- Designed a weighted ranking framework using 6+ workforce parameters to compare company performance and employee satisfaction across industries.
- Derived analytical insights showing 35%+ hiring in IT sectors, 60%+ company presence in metro cities, 85%+ employee job security, and 70%+ low promotion trends.

CERTIFICATES

- Certification in Machine Learning
- Certification in Python Programming
- Certification in Power BI
- Certification in MySQL
- Certification in Exploratory Data Analysis

COURSES

Data Science 2025 – Present
Innomatics Research Labs Hyderabad

LANGUAGES

- English
- Telugu
- Hindi